Paleomagnetism of Early Cambrian redbeds on Cape Breton Island, Nova Scotia

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Received June 21, 1985

Revision accepted March 12, 1986

Results are reported from steeply dipping red sandstones collected at nine sites (45 samples) of the Lower Cambrian Morrison River and MacCodrum formations on southeast Cape Breton Island, which forms part of the Avalon terrane. Demagnetization studies indicate minimal overlap of stability spectra in thermal, alternating-field, and chemical treatments. A stable component (hematite) with high unblocking temperatures (680°C) is dominant at all sites. Its mean direction corrected for geologic tilt is $D = 281°$, $I = +32°$, $k = 97$, giving a paleomagnetic pole (pole M) at 20°N, 146°W (dp, dm = 3°, 6°). The corresponding *in situ* direction is $D = 147°$, $I = +59°$, $k = 97$, with a pole M′ at 2°S, 39°W (dp, dm = 6°, 8°). Poles M and M′ were compared with published Paleozoic poles for the Avalon Zone and cratonic North America to test whether the magnetization originated before or after the Taconic (Ordovician) folding. It could have been acquired during deposition or early diagenesis (Early to Middle Cambrian), in which case the paleolatitude differences between Morrison River – MacCodrum and age-equivalent cratonic sites require a minimum 20° post-Early Cambrian displacement between Avalon Zone and craton. This is, however, difficult to verify, since a fold test is not possible and well-dated Cambrian poles from Avalonia are lacking. Three or four of the many post-Cambrian results from the Avalon and Gander zones were found to be comparable with pole M′, so that alternatively the Morrison River – MacCodrum magnetization could be a postfolding, Late Ordovician – Early Silurian or Late Devonian overprint.

Nous présentons les résultats d'une étude paléomagnétique de 45 échantillons (neuf emplacements) prélevés dans les grès rouges à pendage fortement incliné des formations de Morrison River et de MacCodrum du Cambrien inférieur dans le sud-est de l'île du Cap Breton, lesquels représentent une partie du terrane d'Avalon. Les études de la désaimantation n'indiquent qu'un faible chevauchement des spectres de stabilité dans les traitements thermique, chimique ou par champ alternatif. Une composante stable (hématite) avec une température élevée de déblocage (680°C) domine à tous les empacements échantillonnés. Sa direction moyenne, corrigée pour le basculement géologique, est $D = 281°$, $I = +32°$, $k = 97$, fournissant un pôle paléomagnétique (pôle M) à 20°N, 146°O (dp, dm = 3°, 6°). La direction *in situ* correspondante est $D = 147°$, $I = +59°$, $k = 97$, avec un Pôle M′ à 2°S, 39°O (dp, dm = 6°, 8°). Les pôles M et M′ ont été comparés à d'autres pôles paléolozoïques rapportés ailleurs pour la zone d'Avalon et le craton de l'Amérique du Nord, dans le but de déterminer si l'aimantation fut acquise avant ou après l'orogénie Taconique (Ordovicien). Son acquisition pourrait dater de la période de sédimentation ou de diagenèse précoce (Cambrien inférieur à moyen), et dans un tel cas les différences de paléoaltitude entre le Morrison River – MacCodrum et l'âge équivalent des emplacements cratoniques exigent un déplacement minimum de 20° ultérieurement au Cambrien inférieur entre la zone d'Avalon et le craton. Cependant, ceci est difficile à vérifier car le test de plissement ne peut être appliqué et il n'existe pas d'âge bien défini de pôles cambriens pour l'Avalonie. Trois ou quatre parmi les nombreaux résultats obtenus sur des échantillons post-cambriens sont comparables au pôle M′, il est également possible que l'aimantation des formations Morrison River et MacCodrum soit une surimpression ultérieure à la déformation à l'Ordovicien supérieur – Silurien inférieur ou au Dévonien supérieur.

[Traduit par la revue]

Can. J. Earth Sci. **23**, 1233–1242 (1986)

Introduction

The Avalon Zone

The tectonic evolution of the Avalon Zone (Williams 1978) has been amply reviewed (Haworth and Lefort 1979; Brückner 1980; Piqué 1981). King (1982) suggested that the zone is the result of a tectonic regime transitional between continental margin orogenesis (Rast *et al.* 1976) and intraplate orogenesis (Strong *et al.* 1978). O'Brien *et al.* (1983) regarded the Avalon Zone as a Pan-African terrane that predated the opening of an early Paleozoic proto-Atlantic ocean (Wilson 1966) and became juxtaposed to the Appalachians in the Paleozoic. Possibly the zone evolved as a distinct geologic province or "suspect terrane" (Williams and Hatcher 1982) that may have moved independently from all other Atlantic borderlands. It has not yet proved possible, however, to relocate the Avalon Zone unambiguously on early Paleozoic maps (Williams and Max 1980). Inferred times of motion along the Hermitage Bay – Dover fault system separating the Avalon and Gander zones range from Late Precambrian for the initial faulting (Kennedy *et al.* 1982) to Early Carboniferous for the final fusing of the two zones (Elias and Strong 1982). The Minas Geofracture separating the Meguma and Avalon zones has been attributed (Keppie 1982) to the mid-Devonian Acadian orogeny, the Meguma Zone itself probably being exotic to North America (Schenk 1971).

In principle such drift motions should be extractable from the paleomagnetic record. In practice this has proved difficult, since the times of origin of the remanences being compared across presumed collision sutures or faults are often hard to establish.

Previous paleomagnetic studies on the Avalon Zone

Avalonian paleomagnetic results were first reported on the upper Hadrynian Cabot Group sandstones in Newfoundland

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Printed in Canada / Imprimé au Canada

(Nairn *et al.* 1959) and the Lower Cambrian Ratcliffe Brook sandstones and Upper Devonian or Mississippian Perry lavas and sediments in New Brunswick (Black 1964). In all these studies the techniques employed were insufficient to establish an original field direction. Some newer Avalonian results (Table 2; Fig. 7*a*) represent demonstrably original but not always well-dated magnetizations. A previous Cambrian Cape Breton result (pole CB, Rao *et al.* 1981) is uncertain in age by 60 Ma. The Lower Ordovician Wabana red sandstones (Rao, 1970; Buchan and Hodych 1982, Özdemir and Deutsch 1984) reveal a complex magnetic behaviour: pole WA could be of primary Ordovician or of Devonian age. Two recent Avalonian results with better defined ages are from the latest Ordovician Dunn Point volcanics, Nova Scotia (pole DP, Van der Voo and Johnson 1985), and from the Late Devonian (~360 Ma) St. Lawrence granites, Newfoundland (pole SL, Irving and Strong 1985). Siluro-Devonian results from New Brunswick and Cape Breton (Table 2) fall in two loosely delineated regions comprising (1) poles CB2, JA, and JF and (2) poles CB3, JD, and JE. Group 1 poles, whose boundaries enclose also the somewhat older CB1 and DP poles and two Devonian poles from the Gander Zone of Newfoundland (54°S, 103°E; 4°N, 144°E; Murthy 1983), represent high to intermediate site paleolatitudes. Group 2, with the addition of six late Paleozoic poles (SL, TV, MS, SH, TH, SR; Table 2), is part of the main middle to late Paleozoic North American cluster that includes also most cratonic poles. On an eastern hemisphere plot the group 1 poles lie up to 90° south of that cluster and so have been termed "aberrant"; a possible cause is rapid apparent polar wandering at the sampling sites during Siluro-Devonian time (Roy *et al.* 1979; Rao *et al.* 1981).

Kent and Opdyke (1978, 1979) and Van der Voo *et al.* (1979), observing a 15–20° paleolatitude mismatch between Devonian – Early Carboniferous cratonic sites and some Acadian sites in the northeastern Appalachians, deduced a post-Devonian sinistral displacement of "Acadia" relative to interior North America. Kent (1982) proposed that the Avalon platform shared this hypothetical motion. However, Irving and Strong (1984*a*,*b*, 1985) argued that the magnetizations of cratonic North American rocks crucial to the Acadia displacement hypothesis are dominated by Kiaman (latest Carboniferous and Permian) overprints. From paleomagnetic comparisons between the Avalon Zone and cratonic western Newfoundland they inferred that the two portions of the island had merged by the Early or mid-Carboniferous. It appears now (Kent and Opdyke 1985) that a Carboniferous displacement hypothesis for Acadia is unsupported by paleomagnetism.

More paleomagnetic evidence from the early Paleozoic is needed for constraining past motions of the Avalon Zone. An obvious gap in the record exists in the Cambrian. We report here the first results from the Lower Cambrian Morrison River and MacCodrum formations on Cape Breton Island. This work constitutes a joint project on Paleozoic rocks in the Atlantic Provinces between Laval University and Memorial University of Newfoundland.

Geologic setting

Hutchinson (1952) and Weeks (1954) have studied in detail the stratigraphy and structure of the Mira Valley Cambrian basin (Fig. 1). From their description, the Morrison River Formation, which consists mainly of red sandstones, comprises the oldest Cambrian rocks exposed on the broad Mira Valley syncline. No fossils have been found in this formation, but it grades conformably upwards into red siltstones of the MacCodrum Formation, which contains Early Cambrian trilobites. Overlying the MacCodrum are red and green claystones of the unfossiliferous Canoe Brook Formation, which in turn is overlain by grey to black shales of the Trout Brook Formation of Middle Cambrian age (based on trilobites). The sediments in the basin strike northeasterly, with basal Cambrian rocks exposed continuously on the southeast limb of the syncline, where the beds dip 60–85° W to NW, but occuring over only a short distance on the northwest limb. This structural trend is believed to have been impressed by a deformation near the end of the Ordovician (Taconic orogeny; Hutchinson 1952). The sediments were subsequently intruded by granites of Early to Middle Devonian age during the Acadian orogeny.

Paleomagnetic sampling and measurements

Suitable exposures could be found only on the southeast limb of the syncline where we collected 45 oriented hand samples at nine sites, of which eight are in the Morrison River Formation (M1–M8) and one is in the MacCodrum Formation (MM) (Fig. 1). The direction and moment of natural remanent magnetization (NRM) were measured with a Schonstedt spinner magnetometer (DSM-1); stepwise demagnetizations of NRM were carried out using a Schonstedt alternating-field (AF) demagnetizer (GSD-1) up to 100 mT and a Schonstedt thermal demagnetizer (TSD-1) up to 700°C in residual fields of less than 15 nT. In addition, we employed a two-stage cleaning technique consisting of AF demagnetization followed by thermal demagnetization (Park 1975; Roy and Lapointe 1976), and also chemical leaching (Park 1970; Roy and Park 1974) on a few specimens. Between two and four specimens were used to obtain a sample-mean NRM direction.

The *in situ* site-mean NRM directions are grouped around a mean vector at $D = 156°$, $I = +68°$ (Fig. 2*a*), with intensities ranging from 10^{-2} to 10^{-3} A m^{-1}. The individual directions show no marked concentration near the present Earth's field (PEF) direction at the sampling locality (336°, +71°), indicating the presence of a relatively stable component that could be paleomagnetically significant. The mean NRM direction (156°, +68°) is close to one of three directions characteristic of the Cape Breton "GB" and "GA" type granites (Rao *et al.* 1981), which have radiometric ages between 410 and 460 Ma. For obtaining NRM directions relative to bedding (Fig. 2*b*), a simple correction (tilting the beds about strike into horizontal attitudes) is sufficient because of the relatively simple synclinal structure involved. This rotates the site-mean NRM vector into a shallower, westerly direction (284°, +22°). A fold test is not possible, since only the southeast limb of the syncline could be sampled and the small differences in strike and dip occurring at different sites were insufficient to alter the between-site precision ($k = 10$) through the tilt correction. Application of this tilt correction is meaningful only if the dominant component of the NRM can be assumed to predate the (Late Ordovician?) folding, and this possibility will be discussed later.

AF demagnetization of 45 specimens (one specimen per sample) produced directional changes of 30–40° mostly in declination, along with reductions in intensity of up to 25% with respect to NRM. A very hard magnetization, possibly in hematite, remains at 100 mT as the predominant remanence carrier. This could be the resultant of two or more unresolved hard components with overlapping coercivity spectra. There-

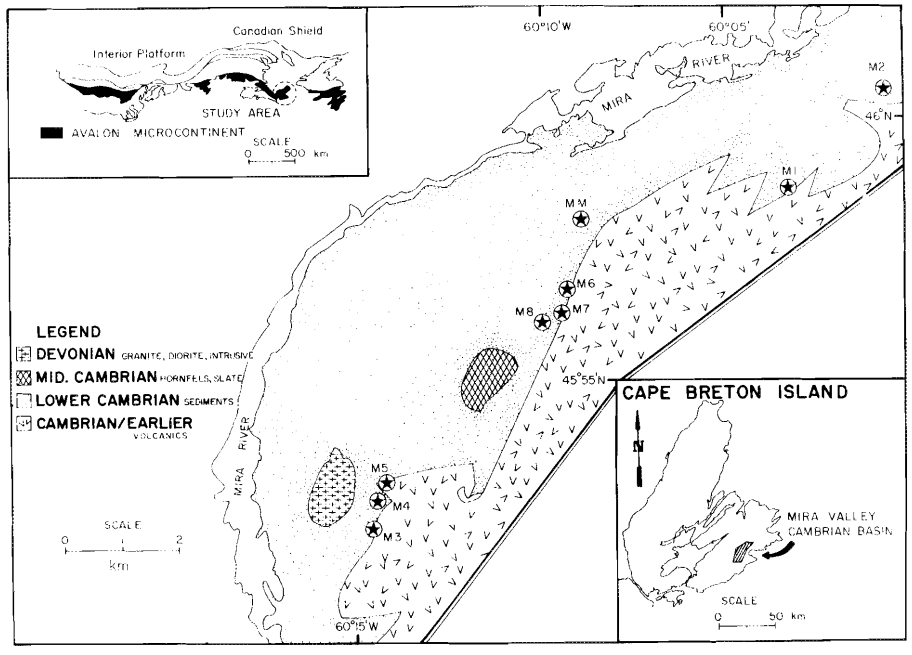


FIG. 1. Simplified geologic map of the Mira Valley basin (after Hutchinson 1952). Stars denote sampling sites in the Morrison River (M1–M8) and MacCodrum (MM) Formations. The inset at upper left shows the extent of the Avalon Zone in the Appalachians (Williams 1978).

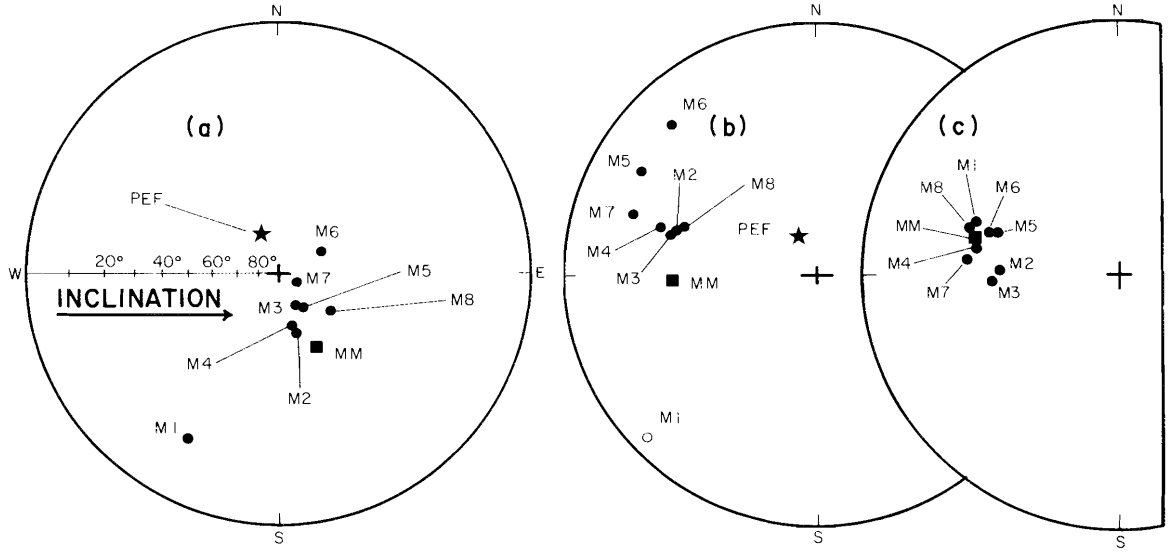


FIG. 2. Site-mean remanence directions of sites M1–M8 and MM (Fig. 1). Natural remanence (*a*) *in situ*, (*b*) -after correction for geologic tilt, (*c*) after stepwise thermal demagnetization (tilt-corrected). Solid circles (M) and squares (MM) represent projections on the lower hemisphere; open circle, upper hemisphere. The star denotes the present Earth's field (PEF) in the sampling region. Wulff equal-area net.

fore, we turned to thermal demagnetization with the aim of distinguishing magnetizations associated with distinct unblocking temperatures (T_{UB}) spectra.

Twenty-seven specimens representing the nine sites were demagnetized in 10 steps to 680°C. Seven specimens gave erratic results and had to be discarded. Results from the remainder were analyzed through orthogonal vector diagrams (Zijderveld 1967). The 20 specimens exhibited a characteristic high-T_{UB} component, which in 14 specimens is evidently the only component present. The other six specimens displayed at least one further component, as shown by two representative examples (Fig. 3). The plots are linear, or nearly so. Constant end points are reached at the high-temperature segment terminating at 680°C, indicating that a stable, single component of magnetization is responsible for the end-point direction. The carrier of this high-T_{UB} remanance is probably hematite, and its tilt-corrected mean direction, calculated from 20 specimens, is 286°, +34°. For the six specimens behaving as in Fig. 3, a further vector subtraction showed that a component with intermediate T_{UB} and a tilt-corrected mean direction 305°, +35° has been removed.

The thermal demagnetization results were checked by subjecting selected specimens to chemical demagnetization. P. Lapointe of the Earth Physics Branch, Ottawa, has kindly

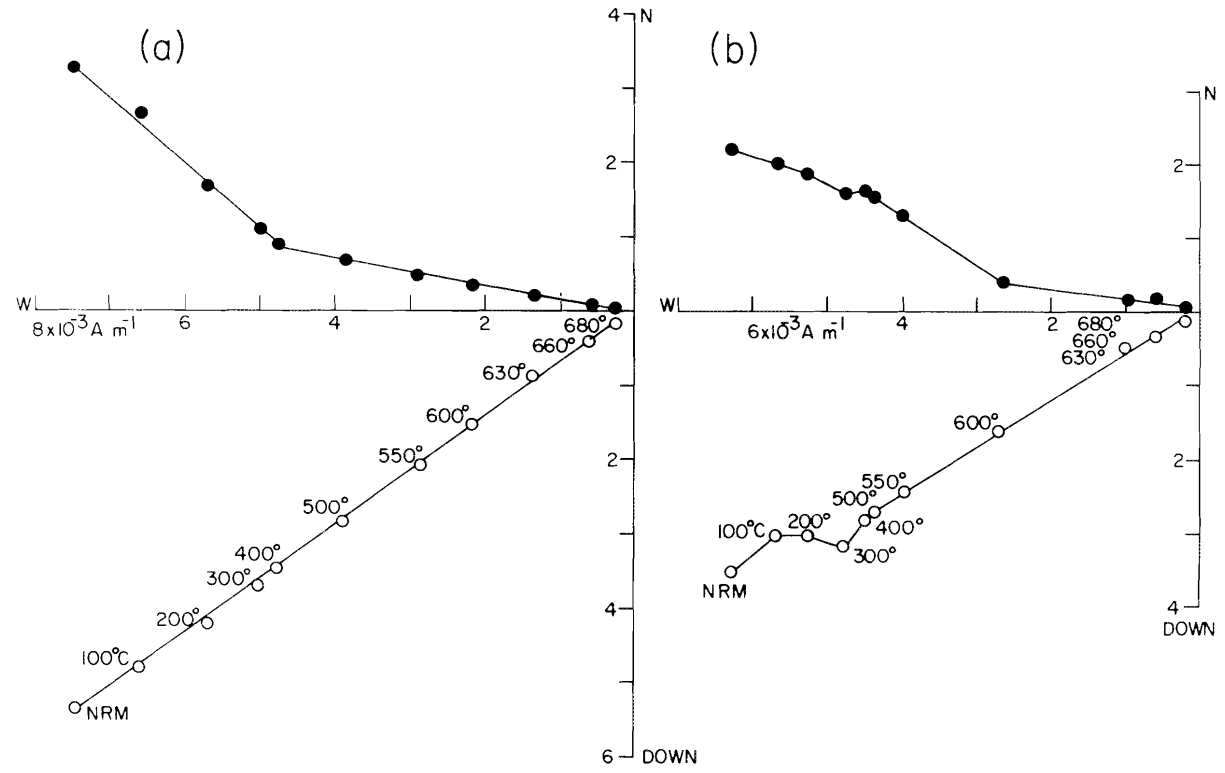


FIG. 3. Orthogonal stepwise thermal demagnetization diagrams (Zijderveld 1967) for two pilot specimens: (*a*) M4-1; (*b*) M7-3. Solid circles represent projections on the horizontal plane; open circles, on the vertical plane. Intensity of magnetization unit along both axes is 10^{-3} A m $^{-1}$.

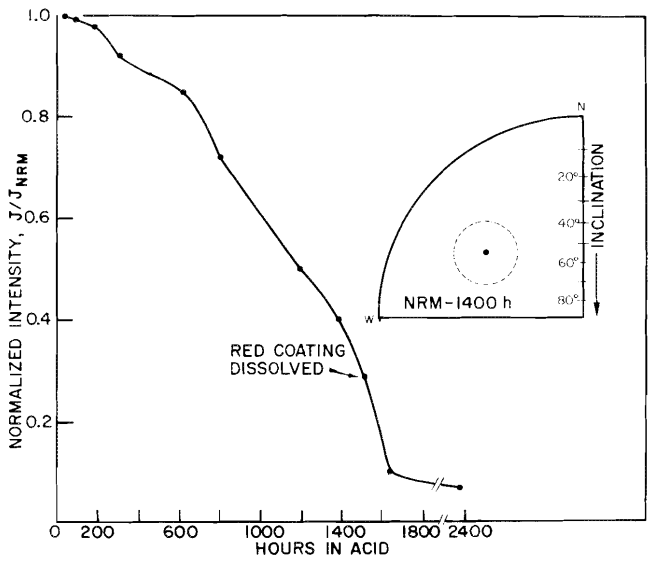


FIG. 4. Decay of normalized intensity (left) and limits of the directional change (dashed circle, right) of specimen M6-3A (J_{NRM} = 6.8 × 10^{-3} A m $^{-1}$) during acid leaching for 1400 h. The dot in the Wulff stereogram is the NRM direction after tilt correction.

performed acid treatment on six fresh specimens, which came from samples that had shown linear Zijderveld plots (Fig. 3). Chemical leaching produced no appreciable deflections of the NRM vector (Fig. 4) during 1400 h of the chemical demagnetization. Further leaching between 1400 and 2300 h tended increasingly to produce random scatter of directions. The chemical results up to 1400 h thus confirm the directional behaviour shown by most specimens in the thermal experiments (Fig. 3)

Although leaching alone can be very effective in the case of red sandstones, we followed it with a two-stage cleaning procedure for isolating possible discrete magnetizations that may elude a one-stage technique. Owing to the weakly cemented nature of the Morrison River sediments, we could not subject many specimens to combined chemical and thermal or AF treatments, and so experiments with AF followed by thermal stepwise demagnetization were performed. Figure 5 shows Zijderveld plots for two of eight specimens that had received AF treatment to 100 mT followed by thermal treatment to 680°C. The plots resemble those based on thermal treatment alone (Fig. 3).

Both the chemical and the dual (AF−thermal) treatments confirm the previous finding from pilot thermal demagnetization of a dominant, hard, high-blocking-temperature component directed westerly and intermediate downward. Among the three treatments (AF, chemical, thermal), when applied singly in the pilot study, the latter was clearly superior in its ability to resolve the stability spectra. Therefore detailed thermal demagnetizations were carried out on the remaining samples for the purpose of calculating Fisher statistics. The high-temperature remanence directions were computed from Zijderveld plots, giving the site-mean value shown in Table 1 and Fig. 2*c*, with an overall mean direction 281°, +32° (N = 9 sites). The site vector grouping (k = 97) is much improved compared with NRM (k = 10).

Magnetic mineralogy

Next we measured the buildup of isothermal remanence (IRM) and the coercivity of remanence (H_{cr}) at room tempera-

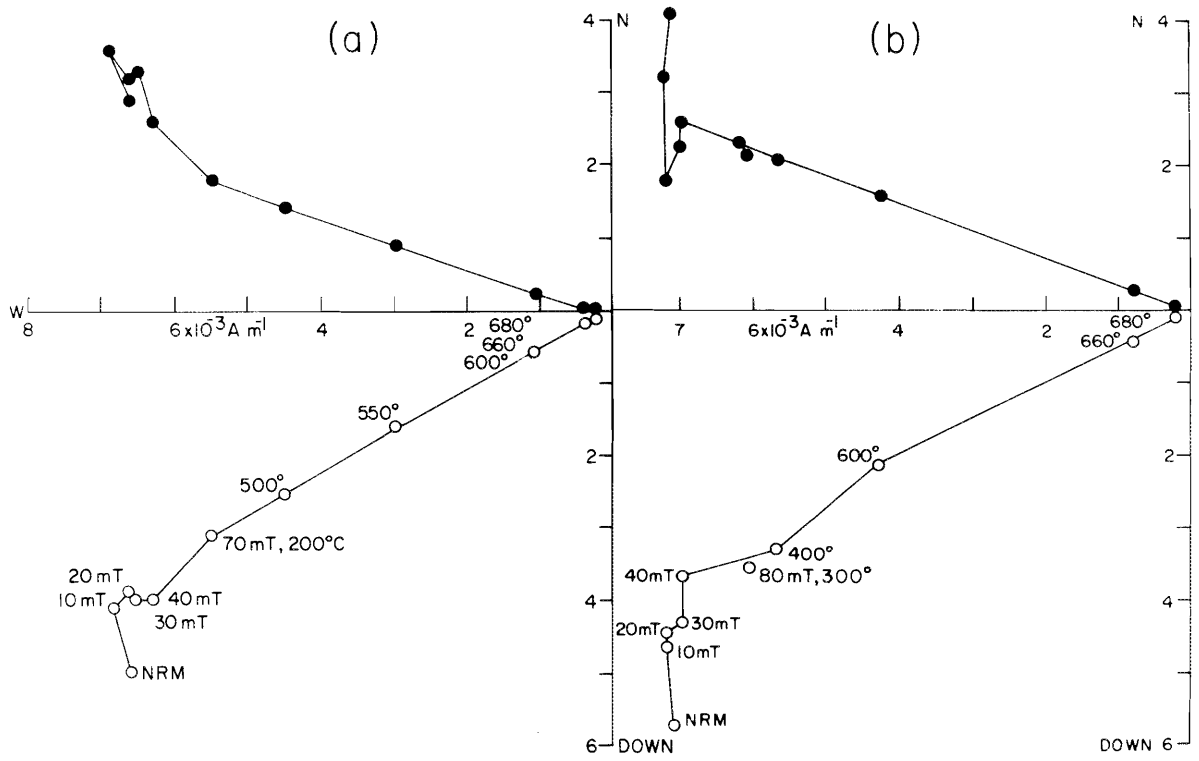


FIG. 5. Results of combined AF−thermal demagnetization for (*a*) specimen M3-1 and (*b*) specimen M7-2, plotted on orthogonal vector diagrams. Symbols and units are as in Fig. 3. Note that for both specimens the final AF step was 100 mT and the first thermal step was 100°C. Since only minimal direction changes occurred in the intervals 70 mT to 200°C (*a*) and 80 mT to 300°C (*b*), the intermediate steps (100 mT, 100°C) were omitted from the notation.

ture in fields up to 1.4 T, in an attempt to identify magnetic minerals. We selected six representative specimens on the basis of characteristic responses to AF and thermal treatment previously shown by other specimens from the same samples. In four of the six specimens the IRM acquisition curves did not reach saturation (Fig. 6*a*), which would be typical of hematite, and the corresponding reverse-field IRM curves yielded high values of H_{cr} (~0.6 T), typical of hematite in red pigment form (Dunlop 1972). The other two specimens reached saturation at fields between 1.0 and 1.4 T, and their reverse-field IRM curve changes slope between two straight-line segments (Fig. 6*b*). This may indicate the presence of two components with different H_{cr} spectra. Following Park (1975), the corresponding values of maximum coercivity of remanence, h_1 = 0.02 T and h_4 = 0.58 mT, were calculated for the two components from the respective apparent H_{cr} values h_2 and h_3. The calculated values (h_1 and h_4) fall, respectively, within the ranges of H_{cr} values reported (Dunlop 1972) for magnetite (0−0.1 T) and pigment hematite (0.3−1.8T). This confirms magnetite and hematite as the likely carriers of remanence in these sediments.

These inferences are confirmed by the optical examination of five polished sections from three and two samples of the Morrison River and MacCodrum formations, respectively, which indicated the presence of fine-grained hematite and minor amounts of (titano?) magnetite.

Analysis of paleomagnetic results

The thermal cleaning (Table 1) resulted in a moderate to large decrease of within-site dispersions and an order of mag-

TABLE 1. Site-mean remanence directions after thermal demagnetization and tilt correction

Site	$N(n)^a$	T_{UB} range (°C)[b]	$D°$, $I°$[c]	k[d]	$\alpha_{95}°$
		MacCodrum Formation (MM)			
MM	5(10)	300−650	285, +28	211	5.0
		Morrison River Formation (M)			
M1	4(12)	400−650	288, +28	44	14.0
M2	4(8)	300−650	271, +39	472	4.2
M3	5(8)	200−630	267, +37	188	5.6
M4	5(10)	500−650	280, +30	37	12.8
M5	5(9)	300−630	289, +35	30	14.2
M6	5(10)	100−650	286, +34	27	15.1
M7	4(8)	400−650	276, +28	61	11.9
M8	4(6)	200−650	288, +26	97	6.8
Total	9 sites	Tilt corrected:	281, +32	97	5.2
		(*In situ*:	147, +59	97	5.2)
		(NRM, *in situ*:	156, +68	10	17.2)

[a] N is the number of samples collected and used in statistics; n is the number of specimens measured per site, usually two or three specimens averaged to obtain a sample direction.

[b] Range in unblocking temperature over which the mean remanence directions are calculated using linear trajectories on vector diagrams such as Figs. 3 and 5.

[c] D and I are the declination and inclination of natural remanent magnetization. Tilt corrections to restore bedding planes are different at each site because of small between-site variations in fold geometry (range of strikes, 195−215°; range of most dips, 70−90°NW).

[d] k is Fisher's precision parameter.

[e] α_{95} is the semi-angle of the cone of 95% confidence for the mean direction.

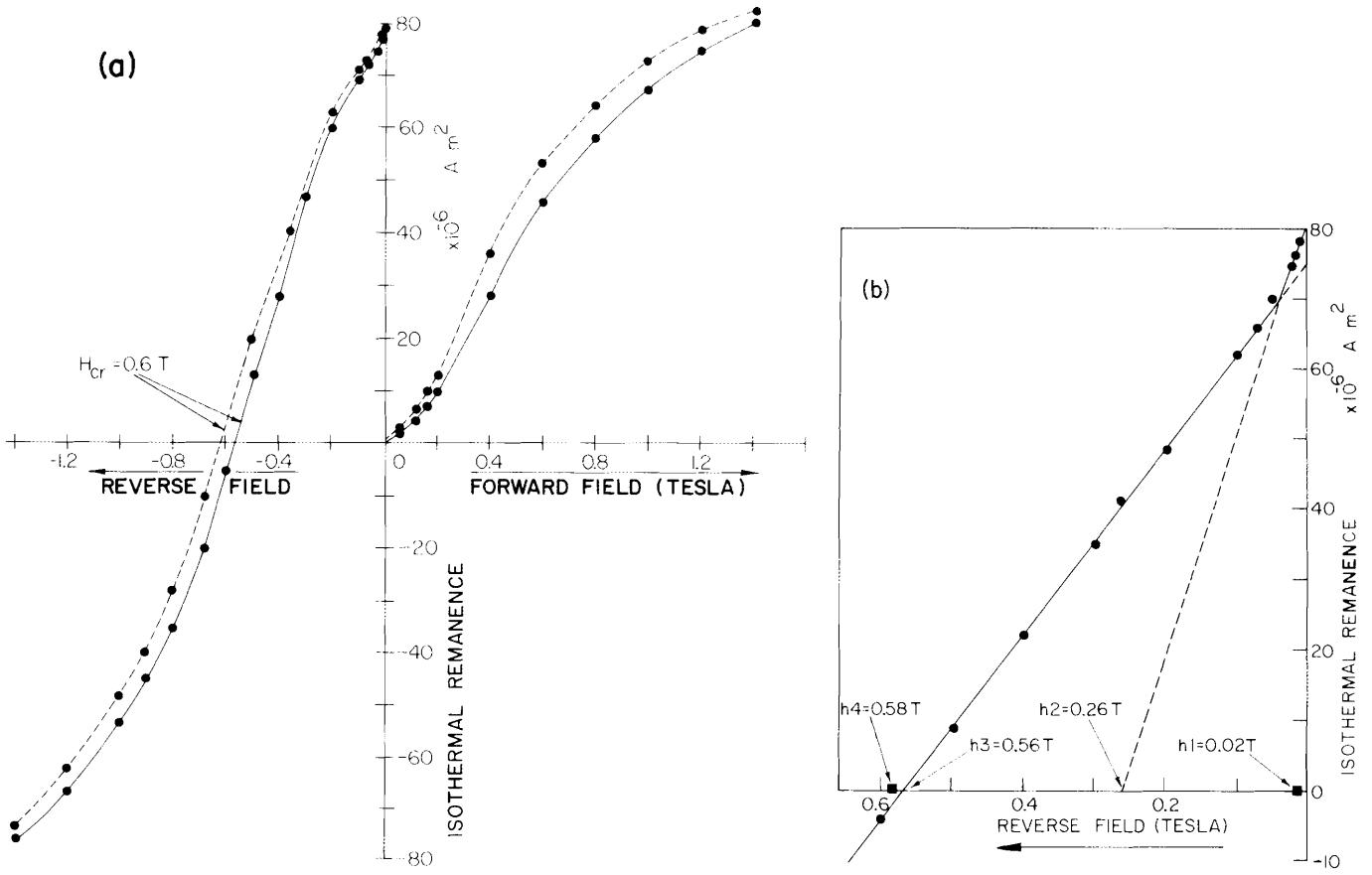


FIG. 6. (*a*) Curves of moment of isothermal remanence (IRM) at room temperature, obtained in forward and reverse fields to 1.4 T, for one Morrison River (solid curve, M8-3A) and one MacCodrum (dashed curve, MM5B) specimen. Intersection of the curves with the reverse-field axis gives the approximate coercivity of remanence $H_{cr} \sim 0.6$ T). (*b*) Two-component IRM curves for one Morrison River specimen (M1-4). H_{cr} values (h_1 and h_4) are computed for the two components from the observed values (h_2 and h_3). Notations as in Park (1975).

nitude decrease of the between-site dispersion, so the treatment was quite effective. The high-blocking-temperature (<680°) remanence direction (281°, +32°, tilt-corrected) dominates the paleomagnetism of the sandstones. The other, relatively "soft" component (magnetite?), associated with the "intermediate"-blocking temperature and low-coercivity spectra, is found only in 10 samples from four of the sites. This component, with a mean *in situ* direction of about 100°, +55°, may be an overprint possibly acquired long after the rocks were formed.

The high-T_{UB} direction is univectorial. It has proved stable against temperatures up to 680°C, against fairly strong fields (100 mT), and against extensive chemical action (1400 h). When plotted, the results of these treatments do not fall along paths of remagnetization circles; however, we cannot rule out that the high-T_{UB} component may be either a thermal overprint acquired through moderate heating over a long time (Pullaiah *et al.* 1975) or even a complete chemical remagnetization occurring long after deposition (Hodych *et al.* 1984). Still, theory and experimental data (e.g., Stacey and Banerjee 1974) show clearly that a stable chemical remanent magnetization (CRM) can be produced in sediments in a geologically short time after deposition, and our laboratory results provide no compelling evidence that these rocks have been substantially overprinted. It is possible, therefore, that the high-temperature component is contemporaneous with deposition or with early diagenesis and so is Early to Middle Cambrian in age. Again, proof is lacking: the Taconic orogeny was the main deformational event affecting the Mira Valley basin (Hutchinson 1952), and because a fold test is not possible, a claim that the stable remanence was acquired before Late Ordovician folding associated with the Taconic orogeny cannot be made conclusively. Following Turner's (1980) paleomagnetic classification of redbeds, our high-temperature component may correspond to his "type C" remanence, which is typified by well-grouped vectors representing essentially a single component postdating the original deposition, though not necessarily postdating early diagenesis, of the sediments.

Interpretation and discussion

Avalon paleomagnetic poles

The tilt-corrected, high-temperature remanence direction inferred at the nine Morrison River – MacCodrum sites (281°, +32°, α_{95} = 5.2°, Table 1) corresponds to a paleomagnetic north pole at *20°N, 146°W; dp, dm = 3°, 6°* (pole M). In citing this pole, we assumed that the remanence originated before (Taconic) tilting: the case for a post-tilting origin is discussed later. In Table 2, pole M is compared with other Paleozoic pole positions reported from the Avalon Zone. In Fig. 7*a* the far-sided antipole (20°S, 34°E) of M is being compared for convenience with a western hemisphere plot of the other Avalon poles and of some representative North American cratonic poles. Most poles in Fig. 7*a* are near sided,

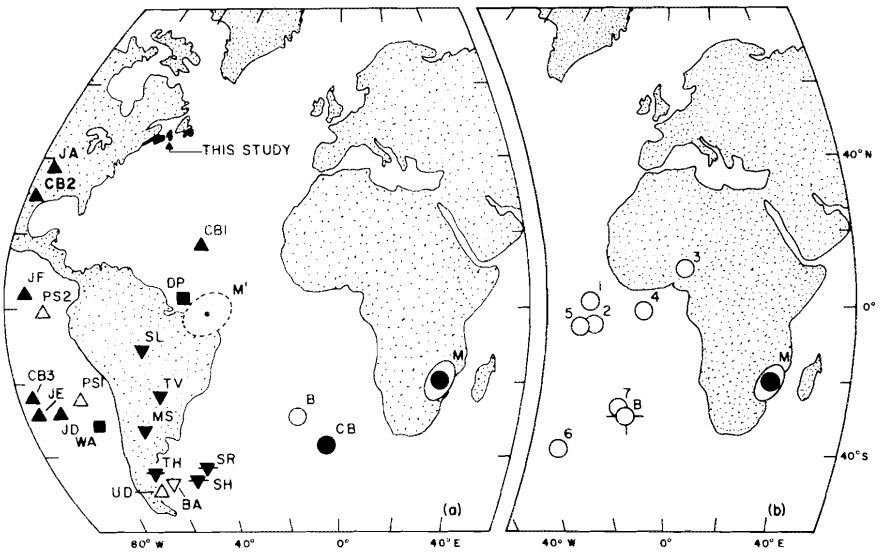


FIG. 7. (*a*) Published paleomagnetic poles from Paleozoic rock units in the Avalon Zone (solid symbols) compared with selected North American cratonic poles (open symbols). Geologic rock ages are denoted by circles (Cambrian), squares (Ordovician), upright triangles (Silurian to mid-Devonian), and inverted triangles (Late Devonian to Early Carboniferous). Inverted triangles with bars denotes poles from mainly Carboniferous rocks cited as probably Kiaman (Late Carboniferous to Early Permian) magnetizations. *Avalon poles*: M, high-temperature pole from the present study after tilt correction, shown with 95% confidence oval; M′ with dashed oval, corresponding *in situ* pole; for other Avalon Zone results, see Table 2. Omitted are poles based on NRM or insufficient stability treatment and of low precision or uncertain geologic or magnetization age. *Cratonic poles*: B, Early Cambrian Bradore Formation, Newfoundland (Rao and Deutsch 1976); PS1, PS2, Early Devonian Peel Sound Formation, Canadian arctic (Dankers 1982); UD, craton mean of Late Devonian poles (Van der Voo and Scotese 1981); BA, Early Carboniferous Barnett Formation, Texas (Kent and Opdyke 1979). BA and UD may actually correspond to Kiaman field directions (Irving and Strong 1984*b*, 1985). (*b*) Pole M is compared with published Early to mid-Cambrian results for the North American craton (open circles): for B, see (*a*). 1, Wood Canyon Formation, Nevada (Gillett and Van Alstine 1979); 2, Tapeats Formation, Arizona (Elston and Bressler 1977); 3–5, Buckingham volcanics, Quebec, groups I, II, III (Dankers and Lapointe 1981); 6, Rome Formation, Virginia–Tennessee (Watts *et al.* 1980); 7, Waynesboro Formation, Virginia–Maryland (Watts *et al.* 1980). Modified cylindrical projection.

and the post-Ordovician ones probably are all geographic south poles, since they are antipodal to the western Pacific region where the middle and late Paleozoic portions of the standard northern apparent polar wandering (APW) curve for cratonic North America are located. The polarity of the three Cambrian poles in Fig. 7*a* (M, CB, B) is less certain because of the far-flung distribution of North American Cambrian poles. The polarity of pole M especially is in doubt because of its wide separation from the cratonic Cambrian poles (Fig. 7*b*).

Let us look at three alternative possibilities for the age of the high-temperature remanence.

Case 1

In this case the remanence was acquired during deposition or early diagenesis. We then note that the tilt-corrected Morrison River magnetic north pole (not shown) and its antipole M both lie far from all post-Cambrian poles in Figs. 7*a*, so that considerable apparent polar wandering must have occurred relative to the Avalon Zone between Early Cambrian and mid Paleozoic time. We now compare pole M with published age-equivalent results for cratonic North America. Eight pole positions with cited Early to early Middle Cambrian magnetization ages are plotted in Fig. 7*b*. All lie significantly west of pole M: five poles are clustered about the equator, and the other three are at latitudes 28–39°S. Hypothetical displacements of the Avalon Zone relative to the craton are best tested (Deutsch 1969) by comparing data from rocks of similar age juxtaposed across the presumed collision suture or fault system. In the case of Morrison River, the only result available for such a comparison is pole B from the Bradore Formation on the Western Platform of Newfoundland, which is part of the craton; this pole is identical with pole 7 for the Waynesboro Formation in Virginia and Maryland. The paleolatitudes calculated from the paleoinclinations at Morrison River (M) and Bradore (B) are 17½°S (or N) and 2°N (or S), respectively. The 20° difference is statistically significant, and from the previous discussion, we believe it is due to plate motion rather than the apparent polar wandering recorded at fixed sites. Then, if case 1 applies, the angular displacement of the Avalon Zone relative to the North American craton must have been at least 20° since the Early Cambrian.

The other Cambrian result for the Avalon Zone (Fig. 7*a*) (antipole CB; 37°S, 4°W; Rao *et al.* 1981) is derived from Cape Breton granites with a radiometric age range of 520 – 580 Ma (Cormier 1973), spanning a significant portion of the Cambrian period. Geologic literature on southeast Cape Breton Island (Weeks 1954; Helmstaedt and Tella 1973; Cormier 1973) suggests conflicting field and age relationships between the Early Cambrian sediments and the 520–580 Ma old granites. In view of this uncertainty, poles M and CB may correspond to widely different Cambrian ages of magnetization.

Case 2

A second alternative for the age of the M magnetization is that it originated after early diagenesis but before folding, that is, between post-Early Cambrian and Late Ordovician time. The only three Avalonian results (Table 2; Fig. 7*a*) whose magnetization ages may fall in that range are pole WA (Wabana sandstones, Early Ordovician), pole DP (Dunn Point volcanics, latest Ordovician), and pole CB1 from Cape Breton granites, which has an inferred age range of 460–430 Ma and so could be also Late Ordovician. Pole WA lies near middle to

TABLE 2. Paleomagnetic poles from the Avalon Zone

Rock unit and location	Age[a]	Pole position (°)[b] Lat., long	dp, dm	Symbol[c]	Reference
Morrison River and MacCodrum formations, red sandstones, Cape Breton	Early Cambrian	20N, 146W	3, 6	M	This study
Granites, Cape Breton	580–520 Ma	37N, 176E	2, 3	CB	Rao *et al.* (1981)
Wabana Group, red sandstones, Newfoundland	Early Ordovician	33N, 102E	3, 6	WA	Buchan and Hodych (1982)
Dunn Point volcanics, Nova Scotia	Latest Ordovician	2S, 136E	5, 7	DP	Van der Voo and Johnson (1985)
Granites, Cape Breton	460–430 Ma	16S, 139E	6, 7	CB1	Rao *et al.* (1981)
Granites, Cape Breton	430–400 Ma	32S, 83E	7, 8	CB2	Rao *et al.* (1981)
Granites, Cape Breton	~400 Ma	27N, 84E	6, 11	CB3	Rao *et al.* (1981)
Mascarene Group, greywackes, New Brunswick	~420 Ma	28N, 85E	2, 4	JE	Roy and Anderson (1981)
Mascarene Group, siltstones, New Brunswick	420–400 Ma	5S, 87E	8, 11	JF	Roy and Anderson (1981)
St. George pluton, gabbro, New Brunswick	~390 Ma	38S, 85E	14, 15	JA	Roy *et al.* (1979)
St. George pluton, gabbro, New Brunswick	~390 Ma	29N, 93E	9, 15	JD	Roy *et al.* (1979)
St. Lawrence granite, Newfoundland	360±5 Ma	12N, 120E	9, 13	SL	Irving and Strong (1985)
Terrenceville Formation, red sandstones, Newfoundland	Early Carboniferous	27N, 124E	6, 11	TV	Kent (1982)
Maringouin, Shepody formations, red sandstones, New Brunswick	Early Carboniferous	36N, 122E	2, 3	MS	Roy and Park (1974)
St. Lawrence granite, "H" component	360±5 Ma[d]	47N, 129E	5, 9	SH	Irving and Strong (1985)
Terrenceville Formation, red sandstones, "B" component	Early Carboniferous	44N, 117E	4, 7	TH	Kent (1982)
Spanish Room Formation, Newfoundland	Carboniferous to Permian[d]	43N, 137E	3, 5	SR	Irving and Strong (1985)

[a]Probable geological age of the rock unit as cited by respective authors.
[b]As cited by the respective authors; dp and dm are semi-axes of ovals of 95% confidence.
[c]Pole symbols are used as in Fig. 7.
[d]Probable magnetization ages have been cited as Late Carboniferous to Permian (Kiaman) by respective authors.

late Paleozoic Avalonian poles, far from pole M. Alternatively, pole WA may represent a post-Ordovician overprint (Buchan and Hodych 1982). Poles DP and CB1 also lie far from pole M, but it is uncertain whether the DP and CB1 magnetizations, assuming they are Late Ordovician, were acquired before or after the presumed Taconic folding at Morrison River. Comparing poles DP and CB1 with the tilt-corrected pole M is justified only in the former case. For these reasons, an unambiguous test of case 2 is unavailable.

Case 3

Here the M magnetization is postfolding. For testing this, one must compare the (thermally treated) *in situ* rather than bedding-related remanence direction with Avalonian results of suitable age. This direction is 147°, +59° (Table 1) and corresponds to a pole M′ at 2°S, 39°W (dp, dm = 6°, 8°), shown with a dashed oval in Fig. 7*a*. The paleolatitude is λ_p = 40°S ± 6°, assuming that M′ is a geographic south pole. For this test to be positive, one would expect pole M′ or its antipole to lie close to the cluster of post-Ordovician Avalonian poles, since a favoured age of overprinting in the Avalon Zone is Devonian (Dallmeyer *et al.* 1983). In fact, pole M′ deviates considerably from nearly all the Silurian–Devonian and Carboniferous poles. So, in general, the postfolding test is negative.

In a few instances, the test may lead to a different conclusion. Let us first compare our result with two Late Devonian results from Newfoundland: pole M′ is only marginally different from pole SL based on the St. Lawrence granite (Irving and Strong 1985). Secondly, pole M′ is statistically identical with a result (not shown in Fig. 7) from the Bonavista Bay dikes (pole BB2: 4°S, 36°W; dp, dm = 9°, 13°; Murthy 1983), which are, however, part of the Gander (not Avalon) Zone. Finally, we look again at the Late Ordovician (or slightly younger) Dunn Point volcanics (DP) and Cape Breton granite (CB1) poles previously considered under case 2. In Fig. 7*a*, both of these poles are seen to lie much closer to M′ than to the tilt-corrected pole M; in fact pole M′ is identical with pole DP, though it is distinct from pole CB1. The paleolatitudes corresponding to poles, SL, BB2, DP, and CB1 are 31°S, 34°S, 42°S, and 55°S, respectively; the first three values at least are not greatly different from λ_p at Morrison River (40°S).

The above comparison suggests the possibility that the

Morrison River magnetization is of secondary, postfolding origin, in which case the probable age of overprinting would be either Late Ordovician–Early Silurian ("Taconic"?) or Late Devonian. Our previous discussion suggests that, alternatively, the Morrison River magnetization may have been acquired during or shortly after deposition of the rocks (case 1). At present, it does not seem possible to choose between these alternatives, that is, between an original versus postfolding age of the remanence.

Acknowledgments

We are grateful to Mr. P. Lapointe of the Earth Physics Branch, Ottawa, for conducting the chemical demagnetization experiments reported in the paper. We also wish to thank Dr. E. Irving, Pacific Geoscience Centre, Dr. R. Van der Voo, University of Michigan, and an unnamed reviewer for their detailed commentaries, which led to several improvements in the manuscript. This work was supported by operating grants from the Natural Sciences and Engineering Research Council of Canada to M. K. Seguin (A-7070) and E. R. Deutsch (A-1946).

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